Linkage isomers:

This type of isomers arises when an ambidentate ligand is bonded to the central metal atom/ion through either of its two different donor atoms. In the below mentioned examples, the nitrite ion is bound to the central metal ion Co^{3+} through a nitrogen atom in one complex,and through oxygen atom in other complex.

$[Co(NH_3)_5(NO_2)]^{2+}$

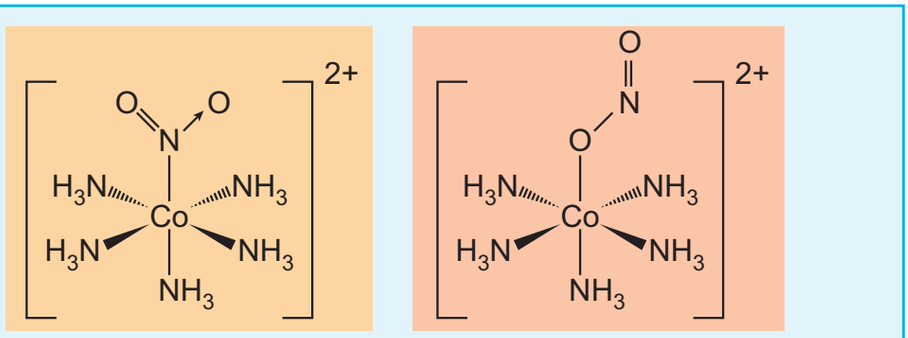


Figure 5.3 Linkage isomers

Coordination isomers:

This type of isomers arises in the coordination compounds having both the cation and anion as complex ions. The interchange of one or more ligands between the cationic and the anionic coordination entities result in different isomers.

For example, in the coordination compound, $[Co(NH_3)_6][Cr(CN)_6]$ the ligands ammonia and cyanide were bound respectively to cobalt and chromium while in its coordination isomer $[Cr(NH_3)_6][Co(CN)_6]$ they are reversed.

Some more examples for coordination isomers

1. $[Cr(NH_3)_5CN][Co(NH_3)(CN)_5]$ and $[Co(NH_3)_5CN][Cr(NH_3)(CN)_5]$
2. $[Pt(NH_3)_4][Pd(Cl)_4]$ and $[Pd(NH_3)_4][Pt(Cl)_4]$

Ionisation isomers:

This type of isomers arises when an ionisable counter ion (simple ion) itself can act as a ligand. The exchange of such counter ions with one or more ligands in the coordination entity will result in ionisation isomers. These isomers will give different ions in solution. For example, consider the coordination compound $[Pt(en)_2Cl_2]Br_2$. In this compound, both Br^- and Cl^- have the ability to act as a ligand and the exchange of these two ions result in a different isomer $[Pt(en)_2Br_2]Cl_2$. In solution the first compound gives Br^- ions while the later gives Cl^- ions and hence these compounds are called ionisation isomers.

Some more example for the isomers,

1. $[Cr(NH_3)_4ClBr]NO_2$ and $[Cr(NH_3)_4Cl\ NO_2]Br$
2. $[Co(NH_3)_4Br_2]Cl$ and $[Co(NH_3)_4Cl\ Br]Br$

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